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Student questions on optimiz...

Lecture 7.pdf

PDF

You are modeling a line of students waiting at office hours for this course.

Above I have shared lecture material such as slides, notes, or a worksheet. Use that as the main source for everything you do here.

Your job

One. Read the material and identify the key ideas and skills it covers.

Two. Based only on this material and basic background that a typical student in this class would already know, create seven realistic student questions.

The seven questions must follow this pattern.

Two quick clarification questions about definitions, notation, or basic setup.

Three why or how questions that dig into reasoning, assumptions, or connections between ideas.

Two application questions that ask what would happen if we change an assumption, parameter, or example.

Requirements for the questions.

Make them sound like real questions a strong but slightly confused student might ask.

Make sure they span the full range of important ideas in the material.

Do not answer the questions.

Number the questions from 1 to 7.

Here are seven realistic student questions that follow your pattern and draw only on the ideas in *Lecture 7* on Muon, spectral norm updates, semi orthogonal updates, and Newton Schulz iteration.

Lecture 7

The questions hit the exact concepts shown in Lecture 7 like spectral norm updates, semi orthogonal matrices, and Newton Schulz. They sounded realistic for confused students after this lecture. I like how there were 3 different types of questions, clarifications, motivation questions, and altering assumptions.

1. Quick clarification questions

1. I'm a little confused about the goal of the optimizer recipe. When we write the update as minimizing the inner product with the gradient subject to a norm constraint, what exactly does that norm represent in practice?

Lecture 7

2. For the semi orthogonal update where we use $UV^\top$, are U and V always coming from the SVD of the gradient at that step, or from the weight matrix itself? I want to make sure I understand which matrix we are decomposing.

Lecture 7

2. Why or how questions (reasoning and motivation)

3. Why does using semi orthogonal matrices help avoid being dominated by the largest singular value? I understand the note saying it gives a more uniform step, but I don't see intuitively how replacing singular values by 1 achieves that.

Lecture 7

4. The notes say that Muon relies on the idea that "getting the direction approximately correct is good enough." Why is direction more important than the exact magnitude in this optimizer, and under what conditions does that approximation still work well?

Lecture 7

5. The Newton Schulz polynomial $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$ eventually maps singular values to 1 when iterated. How does repeatedly applying this polynomial avoid altering the singular vectors while still reshaping the singular values?

Lecture 7

3. Application questions (changing assumptions or conditions)

6. What happens if the singular values of the gradient are not scaled into $(0, 1)$ before applying Newton Schulz? Would the iteration diverge, or just converge more slowly?

Lecture 7

7. Suppose we used a higher order polynomial for $p(x)$ instead of the cubic one. How might that affect stability and computational cost of Muon, especially when training large models?

Lecture 7

If you want, I can also turn these into a worksheet, create misconceptions, generate exam-style questions, or build an interactive tutor prompt for this lecture.

Sources

My answers:

1. The norm represents the shape of the space of directions that we are looking at.
2. It is always coming from the gradient at that step.
3. If the singular values are not close to 1, then as you exponentiate, the largest singular value is goign to grow much faster and dominate.
4. The direction is more important than the exact magnitude because in general we just want gradient to move towards the lower loss region, and approximating is good enough for us because from empircal results we find that this loss function with this constraint works well.
5. Its becasue when you factor out using SVD, we come and reduce back to the originl matrix but with the singular values altered to approach 1
6. it would diverge because it would exponentiate and go out of control
7. I think it makes it more costly for Muon, but it perhaps goes to 1 quicker.



Now act as the instructor for this course.

You have three jobs.

Job one.

Answer each of the seven questions in at most five sentences.
Use the uploaded material as your main source.
You may add minimal outside background only when truly needed.

Job two.

Compare your answer for each question with my answer in my previous message.
For each question
Say whether my answer is fully correct, partially correct, or incorrect.
In two or three sentences explain what I did well and what I missed or got wrong.
If I missed something important or said something inaccurate, give a corrected or improved version of my answer.

Job three.

Make it clear when you are going beyond the notes.
Any time you use information that is not directly supported by the uploaded material, mark that sentence with the tag EXTRA BACKGROUND at the very beginning of the sentence.
If the material does not provide enough information to answer a question confidently, say Not answerable from the notes and explain what additional information would be needed.



Stay closely grounded in the uploaded material whenever possible.
Be honest about uncertainty instead of guessing.

Thought for 1m 54s >

Question 1

Answer.

In the optimizer recipe we choose a norm and then look for the update ΔW with that norm at most η that makes the inner product with the gradient as negative as possible. In the spectral norm case this means ΔW must lie inside the spectral–norm ball of radius η , and solving that small problem gives an update proportional to $U V^T$ transpose when the gradient has SVD $U \Sigma V^T$ transpose. This is the semi orthogonal update used in the notes.

📄 Lecture 7

Your answer. Partially correct.

You were right that the norm is describing the space of directions we are allowed to move in, but you did not mention the size constraint and the specific choice of spectral norm that leads to $U V^T$ transpose. A more complete version would say that the norm both shapes the allowed directions and limits their size to be at most η , and that with the spectral norm this yields a semi orthogonal update $U V^T$ transpose.



When the model compared my answers to its own, it consistently highlighted where my explanations were missing key details from the slides. It did a great job filling in the gaps of my understanding by supplying the intermediate steps and reasoning that the lecture only hinted at. Overall, the model's feedback made my answers more precise and much closer to the logic shown in the lecture.

Question 2

Although the lecture is handwritten and only covers the high level material, the model expands each idea into clearer, more formal math language. This helps because it fills in the gaps the notes leave implicit.

Answer.
Yes, in the derivation on the recall page the matrices U and V come from the SVD of the gradient $\nabla \ell(W)$ at that step, written as $U \Sigma V^T$. The optimizer solution for the spectral norm ball then gives ΔW proportional to $U V^T$. In the Shampoo style formulas later, G_t is the current gradient, and its SVD again provides U and V when that form is used.  Lecture 7

Your answer. Fully correct.
You correctly identified that U and V always come from the gradient at that step, matching the line “if $\nabla \ell(W)$ equals $U \Sigma V^T$ ” in the notes. A slightly more explicit version could mention that this is how we plug the SVD of the gradient into the optimizer recipe to obtain the semi orthogonal update.

Question 3

Answer.
The notes emphasize that semi orthogonal updates have condition number one and give a uniform step in all directions, so the update is not dominated by the largest singular value. Because all singular values of $U V^T$ are one, the update does not stretch some directions much more than others, unlike an update whose singular values vary widely. That is why this choice avoids the situation where the largest singular value drives the behavior of the update.  Lecture 7

Your answer. Partially correct.
You correctly pointed out that unequal singular values can make the largest one dominate, which matches the motivation in the notes that we do not want domination by the largest singular value. However, the key idea here is not exponentiating those values but replacing them all by one, which sets the condition number to one and yields uniform scaling in every direction. A more accurate answer would say that by making all singular values equal to one, $U V^T$ prevents any single direction from growing faster than the others.

Question 4

Whenever it added knowledge not strictly in the notes, it marked it clearly with “EXTRA BACKGROUND.” This transparency makes it easier to see when the model is genuinely summarizing the slides versus reasoning from outside concepts. It also helps highlight when the model’s expansions are helpful versus unnecessary.

Answer.
The notes say that computing $U V^T$ exactly would require an SVD, which is expensive, so Muon relies on two ideas, one of which is that getting the direction approximately correct is good enough. Muon uses Newton Schulz iterations to approximate the mapping from $U \Sigma V^T$ to $U V^T$ by manipulating the singular values. EXTRA BACKGROUND The idea is that as long as the update points roughly in the descent direction determined by the gradient, small errors in its exact shape can be tolerated while still making progress on the loss.  Lecture 7

Your answer. Partially correct.
You captured the high level idea that the direction matters more than exact magnitude and that empirically this works well, which is consistent with the note that an approximate direction is good enough. What you missed is the specific link to the cost of exact SVD and the role of Newton Schulz as the mechanism that gives that approximate direction efficiently. A refined answer would connect “approximate direction” directly to avoiding full SVD and using the polynomial iterations instead.

Question 5

Answer.
The Newton Schulz section shows that for an odd polynomial p , we have p applied to $U \Sigma V^T$ equals $U p(\Sigma) V^T$. This means applying p to the matrix changes only the singular values, by sending each singular value σ to $p(\sigma)$, and keeps the singular vectors U and V the same. The notes then choose a polynomial p whose iterates drive positive inputs toward one, so repeated application pushes all singular values toward one while preserving singular vectors.  Lecture 7

Your answer. Partially correct.
You correctly said that we end up with the original matrix but with singular values altered to approach one, which is the main idea. However, the crucial step is the identity $p(U \Sigma V^T) = U p(\Sigma) V^T$, which explains precisely why the singular vectors are unchanged. An improved answer would explicitly mention that property and then say that picking p so that $p(x)$ tends to one for x bigger than zero makes the singular values converge to one under iteration.

Question 6

Answer.
The plots and notes state that we need singular values in the interval from zero to one before applying the Newton Schulz polynomial, and they suggest doing this by normalizing the matrix by its Frobenius norm.  Lecture 7 For values outside that interval, the displayed curve shows that p of values drives them away from

 For values outside that interval, the displayed curve shows that p or x moves them away from one rather than toward it, so iterating p would not drive those singular values to one. Thus the orthogonalization idea would fail and could lead to unstable behavior instead of converging to $U V^T$.

Your answer. Partially correct.

You were right that something goes wrong and that the procedure can become uncontrolled if the singular values are not scaled into the required range, which matches the warning in the notes. However, the notes do not describe this as simple exponentiation, and they do not fully spell out the exact failure mode, only that we first need to scale the spectrum into zero to one. A better answer would say that without this scaling the polynomial iteration no longer has the property that repeated application pushes the singular values toward one, so the method can become unstable.

Question 7

Answer.

The Muon summary page explains that we can choose p as a higher order odd polynomial with more terms. It notes that higher order choices may converge faster, in the sense that fewer iterations are needed to push singular values near one, but each evaluation of p is more expensive. Thus there is a tradeoff between convergence speed and per step computational cost when selecting the polynomial.

 Lecture 7

Your answer. Fully correct.

You correctly identified both sides of the tradeoff faster movement of singular values toward one and increased cost for Muon when using a higher order polynomial, which matches the bullets in the notes. A slightly sharper version could state explicitly that the notes say higher order may converge faster but each step is more expensive.

Sources

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It didn't make up definitions or equations, which is good. But sometimes it confidently stated that the notes "say" something that is not necessarily easy to extract. These moments aren't harmful but worth noting.